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Preconditions for Effective Debugging

- In order to avoid excessive time spent on debugging, the programmer should be mentally prepared for the effort. The following steps are useful to prepare for debugging.
- **Understand the design and algorithm** - If you are working on a module and you do not understand its design or its algorithms, then debugging will be very difficult. If you don't understand the design then you can't test the module because you do not know what it is supposed to do. If you don't understand the algorithms then you will find it very difficult to locate the errors that are revealed by testing. A second reason for the importance of understanding algorithms is that you may need that understanding in order to construct good test cases. This is especially true for algorithms for complex data structures.
- **Check correctness** - There are several methods for checking correctness of an implementation prior to execution.
- **Correctness proofs** - One useful code check is to examine code using the logical methods of correctness proofs. For example, if you know preconditions, invariants, terminating conditions, and postconditions for a loop then there are some easy checks that you can make. Does the precondition, together with any loop entry code imply that the invariant is initially true? Does the loop body preserve the invariant? Does execution of the loop body make progress towards loop termination? Does the invariant, together with the loop terminating condition and loop exit code, imply the postcondition? Even if these checks don't find all errors, you will often gain a better understanding of the algorithm by making the checks.

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- **Code tracing** –

- Often, errors can be detected by tracing through the execution of various calls to module services, starting with a variety of initial conditions for the module. For poorly understood psychological reasons, tracing works best if you are describing your tracing to someone else. In order to be effective, tracing of a procedure or function should be done assuming that calls to other procedures and functions work correctly, even if they are recursive calls. If you trace into a called procedure or function then you will find yourself dealing with too many levels of abstraction. This usually leads to confusion. If there is any doubt about the called procedures and functions then they can be traced separately to verify that they perform according to specifications. Again, tracing may not catch all errors, but it can enhance your understanding of algorithms.
- **Anticipate errors** - Unfortunately, humans make errors with correctness arguments and sometimes miss cases in code tracing, and peers don't always catch errors either. So a programmer should be prepared for some errors remaining in the code after the steps listed above. Hopefully, there won't be too many.

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- **Peer reviews –**

- A peer review involves having a peer examine your code for errors. To be effective, the peer should either already be familiar with the algorithm, or should be given the algorithm and code in advance. When the reviewer meets with the code writer, the code writer should present the code with explanations of how it correctly implements the algorithm. If the reviewer doesn't understand or disagrees with part of the implementation, they discuss that part until both are in agreement about whether or not it is an error. The reviewer's role is only as an aid to detecting errors. It is left to the implementor to correct them. Much of the benefit of a peer review derives from the psychology of presenting how something works. Often the code writer discovers his or her own errors during the review. In any case, it is useful to have an outsider review your work in order to get a different perspective and to discover blind spots that seem to be inherent in evaluating your own work. Like code tracing, peer reviews can be time consuming. For class work, a peer review of an entire module is not likely to pay for itself in terms of instructional value. So reviews should be restricted to short segments of code. For commercial programming, however, quality of the code is much more important. Thus peer reviews are a significant part of a software quality assurance program.

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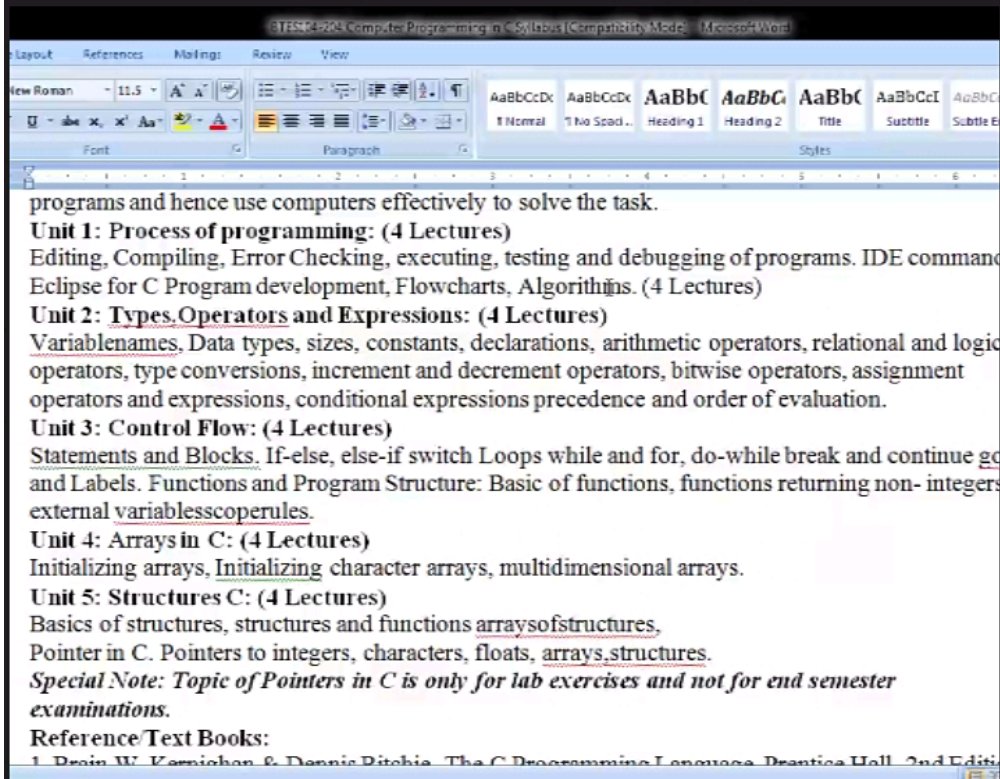


Requirements for Debugging

- To effectively debug a module, it is necessary to have some method for calling upon the services provided by the module. There are two common methods for doing this.
- **Hardwired drivers** - A hardwired driver is a main program module that contains a fixed sequence of calls to the services provided by the module that is being tested. The sequence of calls can be modified by rewriting the driver code and recompiling it. For testing modules whose behavior is determined by a very small number of cases, hardwired drivers offer the advantage of being easy to construct. If there are too many cases, though, they have the shortcoming that a considerable effort is involved in modifying the sequence of calls.
- **Command interpreters** - A command interpreter drives the module under test by reading input and interpreting it as commands to execute calls to module services. Command interpreters can be designed so that the commands can either be entered interactively or read from a file. Interactive command interpretation is often of great value in early stages of debugging, whereas batch mode usually is better for later stages of debugging and final testing. The primary disadvantage of command interpreters is the complexity of writing one, including the possibility that a lot of time can be spent debugging the interpreter code. This is mitigated by the fact that most of the difficult code is reusable, and can be easily adapted for testing different kinds of modules. For almost all data structure modules, the flexibility offered by command interpreters makes them a preferred choice.

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